

Chapter

Probability

3

GLOBAL
EDITION



Elementary Statistics

Picturing the World

EIGHTH EDITION

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Chapter Outline

- 3.1 Basic Concepts of Probability and Counting
- 3.2 Conditional Probability and the Multiplication Rule
- 3.3 The Addition Rule
- 3.4 Additional Topics in Probability and Counting

Section 3.1

Basic Concepts of Probability and Counting

Section 3.1 Objectives

- How to identify the sample space of a probability experiment and how to identify simple events
- How to use the Fundamental Counting Principle to find the number of ways two or more events can occur
- How to distinguish among classical probability, empirical probability, and subjective probability
- How to find the probability of the complement of an event
- How to use a tree diagram and the Fundamental Counting Principle to find probabilities

Probability Experiments

Probability experiment

- An action, or trial, through which specific results (counts, measurements, or responses) are obtained.

Outcome

- The result of a single trial in a probability experiment.

Sample Space

- The set of all possible outcomes of a probability experiment.

Event

- Consists of one or more outcomes and is a subset of the sample space.

Example: Identifying the Sample Space of a Probability Experiment

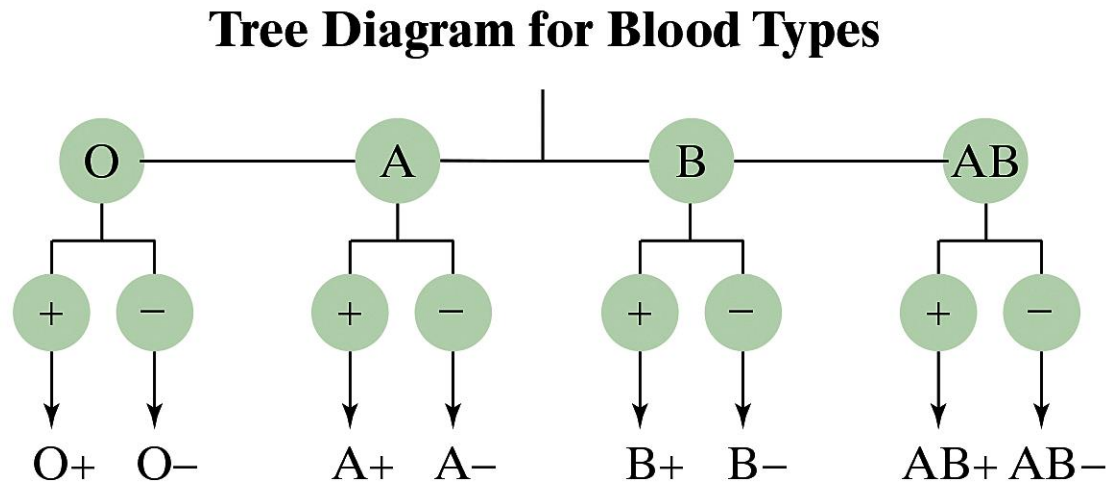
A survey consists of asking people for their blood types (O, A, B, and AB), including whether they are Rh-positive or Rh-negative. Determine the number of outcomes and identify the sample space.

Solution:

There are four blood types: O, A, B, and AB. Each person is either Rh-positive or Rh-negative. A **tree diagram** gives a visual display of the outcomes by using branches that originate from a starting point. It can be used to find the number of possible outcomes in a sample space as well as individual outcomes.

Solution: Identifying the Sample Space of a Probability Experiment

Tree diagram:



The sample space has eight possible outcomes, which are listed below.

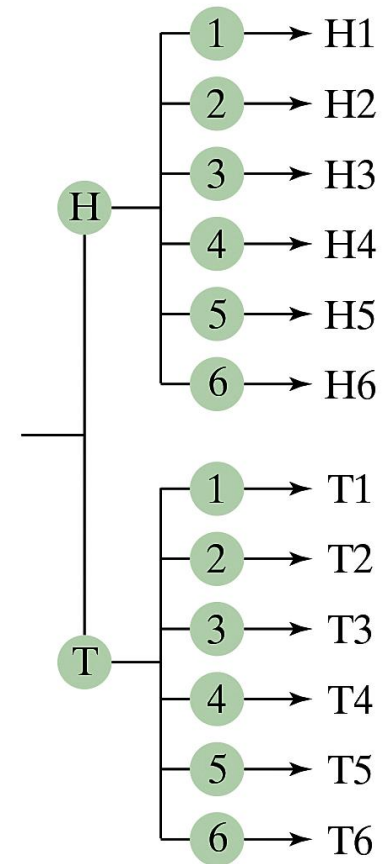
$\{O+, O-, A+, A-, B+, B-, AB+, AB-\}$

Simple Events

Simple event

- An event that consists of a single outcome.
 - e.g. “Tossing heads and rolling a 3”
 $A = \{H3\}$
- An event that consists of more than one outcome is not a simple event.
 - e.g. “Tossing heads and rolling an even number” $B = \{H2, H4, H6\}$

Diagram for Coin and Die Experiment



Example: Identifying Simple Events

Determine the number of outcomes in each event. Then decide whether each event is simple or not. Explain your reasoning.

1. For quality assurance, you randomly select a machine part from a batch that has been manufactured that day. Event A is selecting a specific defective machine part.

Solution:

Event A has only one outcome: choosing the specific defective machine part. **So, the event is a simple event.**

Example: Identifying Simple Events

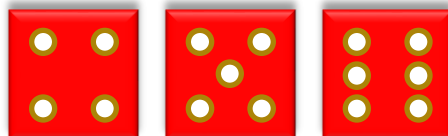
Determine the number of outcomes in each event. Then decide whether each event is simple or not. Explain your reasoning.

2. You roll a six-sided die. Event B is rolling at least a 4.

Solution:

Event B has three outcomes: rolling a 4, a 5, or a 6.

Because the event has more than one outcome, it is **not simple**.



The Fundamental Counting Principle

The Fundamental Counting Principle

- If one event can occur in m ways and a second event can occur in n ways, the number of ways the two events can occur in sequence is $m \cdot n$.
- Can be extended for any number of events occurring in sequence.

Example: Using the Fundamental Counting Principle



You are purchasing a new car. The possible manufacturers, car sizes, and colors are listed in the table.

Manufacturer	Car size	Color
Ford	compact	white (W)
GM	midsize	red (R)
Honda		black (B)
		green (G)

How many different ways can you select one manufacturer, one car size, and one color? Use a tree diagram to check your result.

Solution: Using the Fundamental Counting Principle

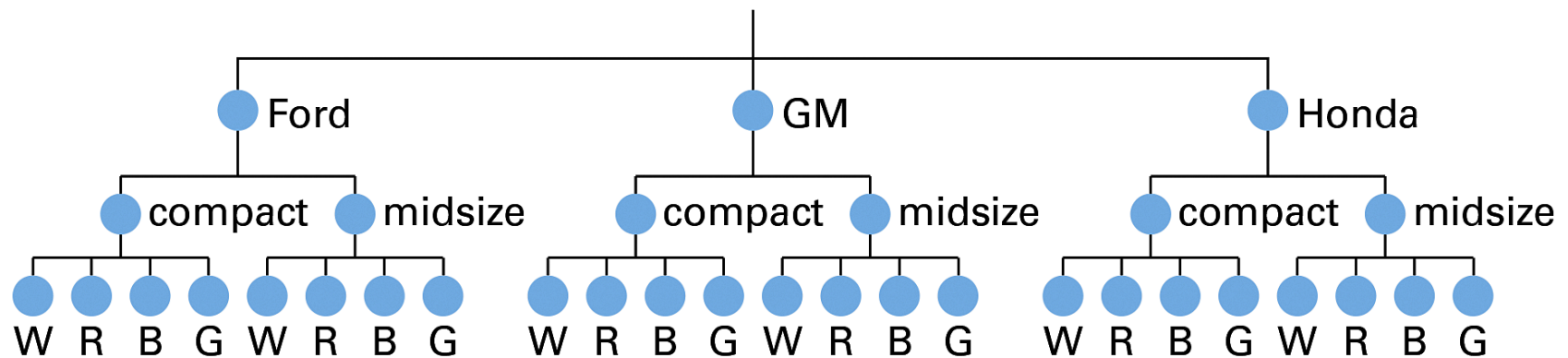


There are three choices of manufacturers, two car sizes, and four colors.

Using the Fundamental Counting Principle:

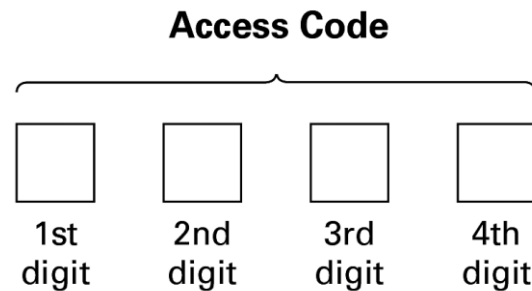
$$3 \cdot 2 \cdot 4 = 24 \text{ ways}$$

Tree Diagram for Car Selections



Example: Using the Fundamental Counting Principle

The access code for a car's security system consists of four digits. Each digit can be any number from 0 through 9.



How many access codes are possible when

1. each digit can be used only once and not repeated?
2. each digit can be repeated?
3. each digit can be repeated but the first digit cannot be 0 or 1?

Solution: Using the Fundamental Counting Principle

Solution

1. Because each digit can be used only once, there are 10 choices for the first digit, 9 choices left for the second digit, 8 choices left for the third digit, and 7 choices left for the fourth digit. Using the Fundamental Counting Principle, you can conclude that there are

$$10 \cdot 9 \cdot 8 \cdot 7 = 5040$$

possible access codes.

Solution: Using the Fundamental Counting Principle

Solution

2. Because each digit can be repeated, there are 10 choices for each of the four digits. So, there are

$$10 \cdot 10 \cdot 10 \cdot 10 = 10^4 = 10,000$$

possible access codes.

3. Because the first digit cannot be 0 or 1, there are 8 choices for the first digit. Then there are 10 choices for each of the other three digits. So, there are

$$8 \cdot 10 \cdot 10 \cdot 10 = 8000$$

possible access codes.

Solution: Using the Fundamental Counting Principle

Solution

Remember that you can use technology to check your answers. For instance, a TI-84 Plus was used to check the results.

TI-84 PLUS	
$10 \times 9 \times 8 \times 7$	5040
10^4	10000
$8 \times 10 \times 10 \times 10$	8000

Types of Probability

Classical (theoretical) Probability

- Each outcome in a sample space is equally likely.
- $$P(E) = \frac{\text{Number of outcomes in event } E}{\text{Number of outcomes in sample space}}$$

Example: Finding Classical Probabilities

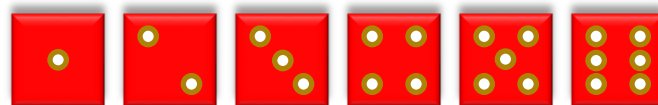
You roll a six-sided die. Find the probability of each event.

1. Event A : rolling a 3
2. Event B : rolling a 7
3. Event C : rolling a number less than 5



Solution:

Sample space: $\{1, 2, 3, 4, 5, 6\}$



Solution: Finding Classical Probabilities

1. Event A : rolling a 3 Event $A = \{3\}$

$$P(\text{rolling a 3}) = \frac{1}{6} \approx 0.167$$



2. Event B : rolling a 7 Event $B = \{ \}$ (7 is not in the sample space)

$$P(\text{rolling a 7}) = \frac{0}{6} = 0$$

3. Event C : rolling a number less than 5

$$\text{Event } C = \{1, 2, 3, 4\}$$

$$P(\text{rolling a number less than 5}) = \frac{4}{6} \approx 0.667$$

Types of Probability

Empirical (statistical) Probability

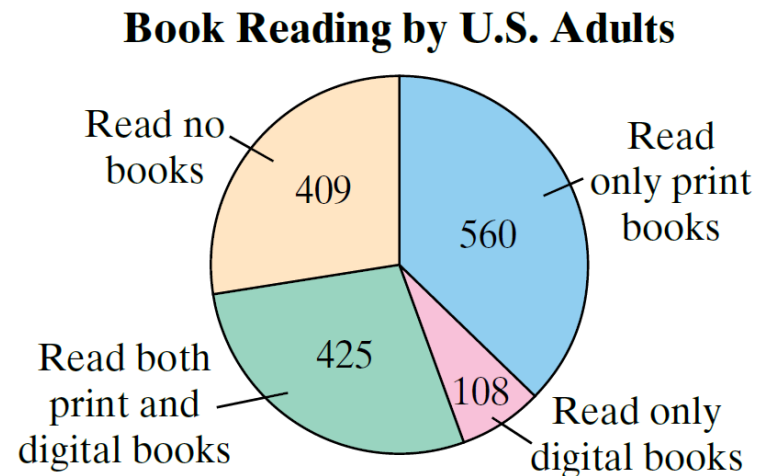
- Based on observations obtained from probability experiments.
- Relative frequency of an event.

- $$P(E) = \frac{\text{Frequency of event } E}{\text{Total frequency}} = \frac{f}{n}$$

Note that $n = \sum f$.

Example: Finding Empirical Probabilities

A company is conducting a survey of randomly selected U.S. adults to determine how they read books during the past year, if at all. So far, 1502 adults have been surveyed. The pie chart shows the results. (Note that digital books include ebooks as well as audio books.) What is the probability that the next adult surveyed read only print books during the last year? (*Adapted from Pew Research Center*)



Solution: Finding Empirical Probabilities

Solution

- Note that the responses are *not* equally likely to occur and are based on observations.
- You *cannot* use the formula for classical probability, but you can use the formula for empirical probability.
- The event is a response of “read only print books.” The frequency of this event is 560. The total of the frequencies is

$$n = 560 + 108 + 425 + 409 = 1502.$$

Solution: Finding Empirical Probabilities

Solution

- The empirical probability that the response of the next adult is “read only print books” is

$$P(\text{read only print books}) = \frac{560}{1502} \approx 0.373.$$

Example: Using a Frequency Distribution to Find Probabilities

A company is conducting a survey of randomly selected individuals to determine the ages of a social media application. So far, 3000 users of the application have been surveyed. The frequency distribution shows the results.

What is the probability that the next user surveyed is 25 to 34 years old? (*Adapted from Statista*)

Ages	Frequency, f
13 to 17	84
18 to 24	459
25 to 34	765
35 to 44	546
45 to 54	432
55 to 64	369
65 and over	345
	$\Sigma f = 3000$

Solution: Using a Frequency Distribution to Find Probabilities

Solution

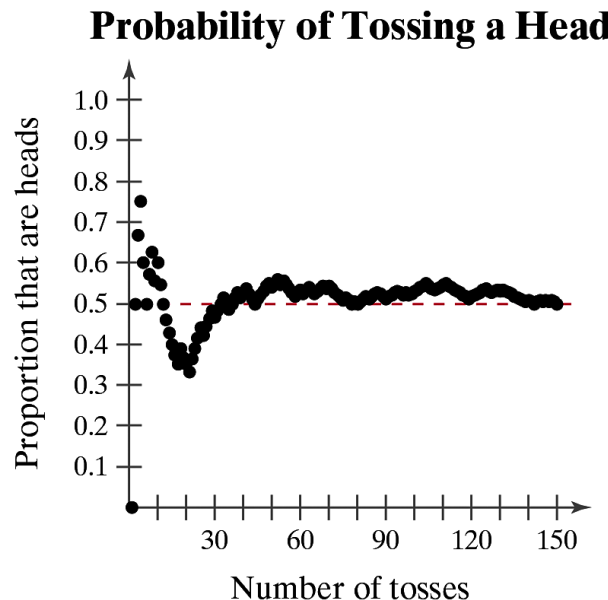
- Because the responses are *not* equally likely to occur and are based on observations, use the formula for empirical probability.
- The event is a response of “25 to 34 years old.” The frequency of this event is 765.
- The total of the frequencies is 3000, the empirical probability that the next user is 25 to 34 years old is

$$P(\text{age 25 to 34}) = \frac{765}{3000} = 0.255.$$

Law of Large Numbers

Law of Large Numbers

- As an experiment is repeated over and over, the empirical probability of an event approaches the theoretical (actual) probability of the event.



Types of Probability

Subjective Probability

- Intuition, educated guesses, and estimates.
- e.g. A doctor may feel a patient has a 90% chance of a full recovery.



Example: Classifying Types of Probability

Classify the statement as an example of classical probability, empirical probability, or subjective probability. Explain your reasoning.

1. The probability that you will get an A on your next test is 0.9.

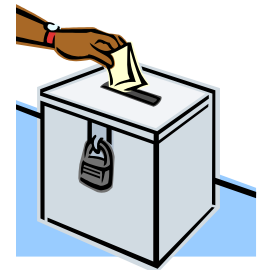
Solution:

Subjective probability (most likely an educated guess)

Example: Classifying Types of Probability

Classify the statement as an example of classical probability, empirical probability, or subjective probability. Explain your reasoning.

2. The probability that a voter chosen at random will be younger than 35 years old is 0.3.



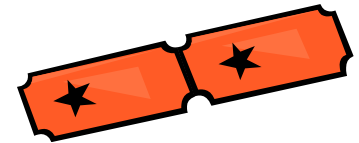
Solution:

Empirical probability (most likely based on a survey)

Example: Classifying Types of Probability

Classify the statement as an example of classical probability, empirical probability, or subjective probability. Explain your reasoning.

3. The probability of winning a 1000-ticket raffle with one ticket is $\frac{1}{1000}$.



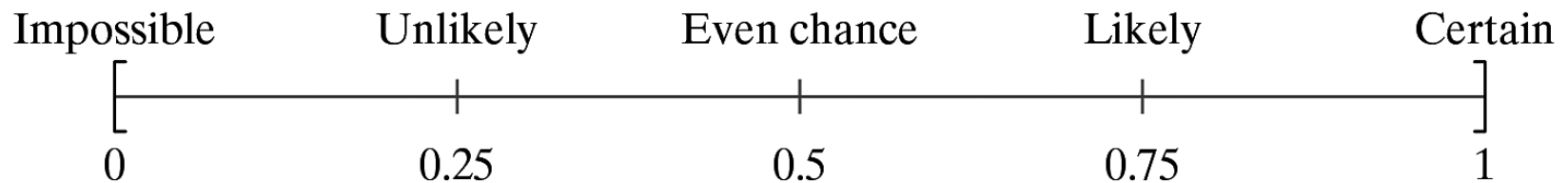
Solution:

Classical probability (equally likely outcomes)

Range of Probabilities Rule

Range of probabilities rule

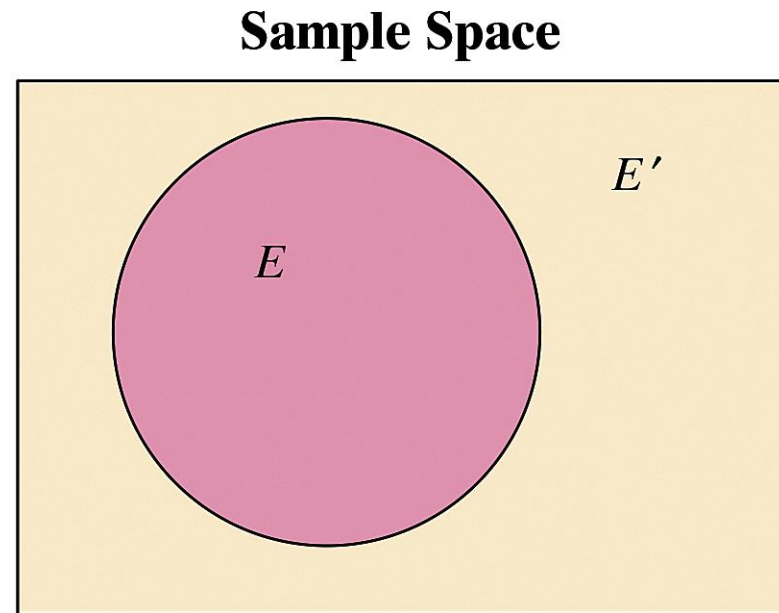
- The probability of an event E is between 0 and 1, inclusive.
- $0 \leq P(E) \leq 1$



Complementary Events

Complement of event E

- The set of all outcomes in a sample space that are not included in event E .
- Denoted E' (E prime)
- $P(E) + P(E') = 1$
- $P(E) = 1 - P(E')$
- $P(E') = 1 - P(E)$



Example: Finding the Probability of the Complement of an Event

Find the probability of randomly selecting a social networking site user who is not 25 to 34 years old.

Ages	Frequency, f
13 to 17	84
18 to 24	459
25 to 34	765
35 to 44	546
45 to 54	432
55 to 64	369
65 and over	345
	$\Sigma f = 3000$

Solution: Finding the Probability of the Complement of an Event

Use empirical probability to find $P(\text{age 25 to 34})$.

$$P(\text{age 25 to 34}) = \frac{765}{3000} = 0.255.$$

Ages	Frequency, f
13 to 17	84
18 to 24	459
25 to 34	765
35 to 44	546
45 to 54	432
55 to 64	369
65 and over	345
	$\Sigma f = 3000$

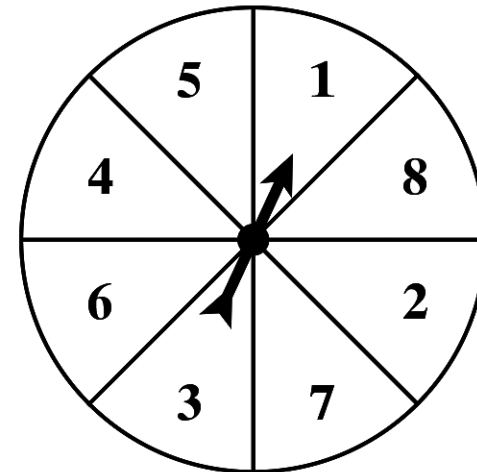
So, the probability that a user is not 25 to 34 years old is

$$P(\text{age is not 25 to 34}) = 1 - \frac{765}{3000} = 0.745.$$

Example: Using a Tree Diagram

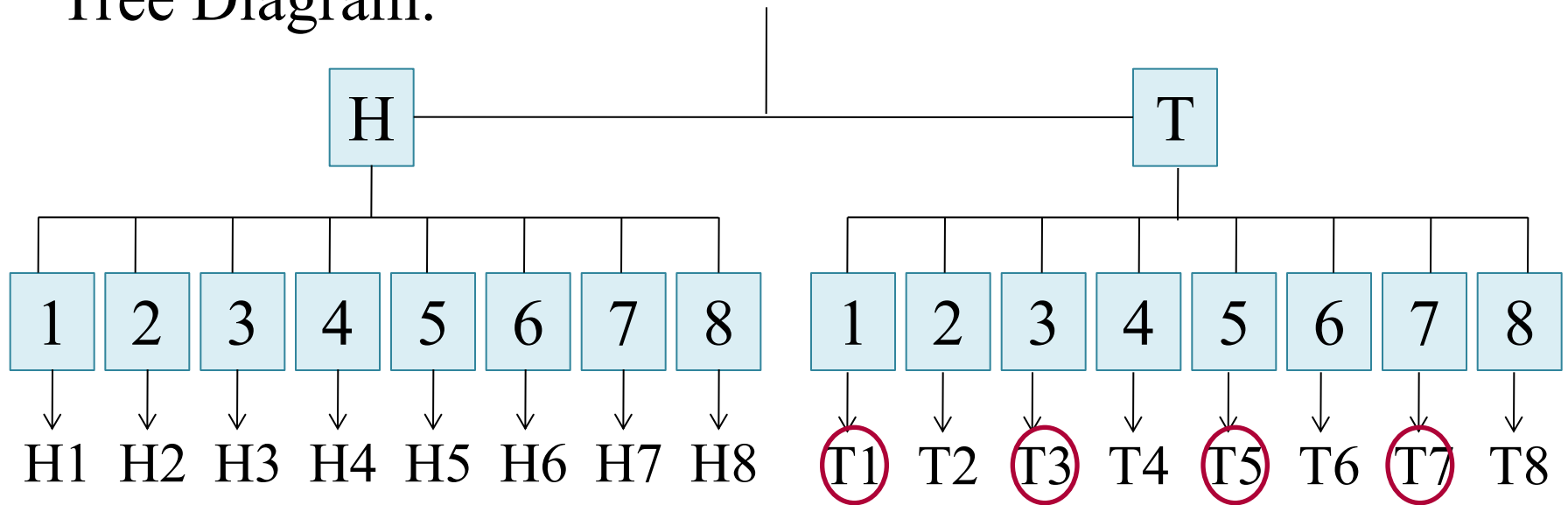
A probability experiment consists of tossing a coin and spinning the spinner shown. The spinner is equally likely to land on each number. Use a tree diagram to find the probability of each event.

1. Event A: tossing a tail and spinning an odd number
2. Event B: tossing a head or spinning a number greater than 3



Solution: Probability Using a Tree Diagram

Tree Diagram:

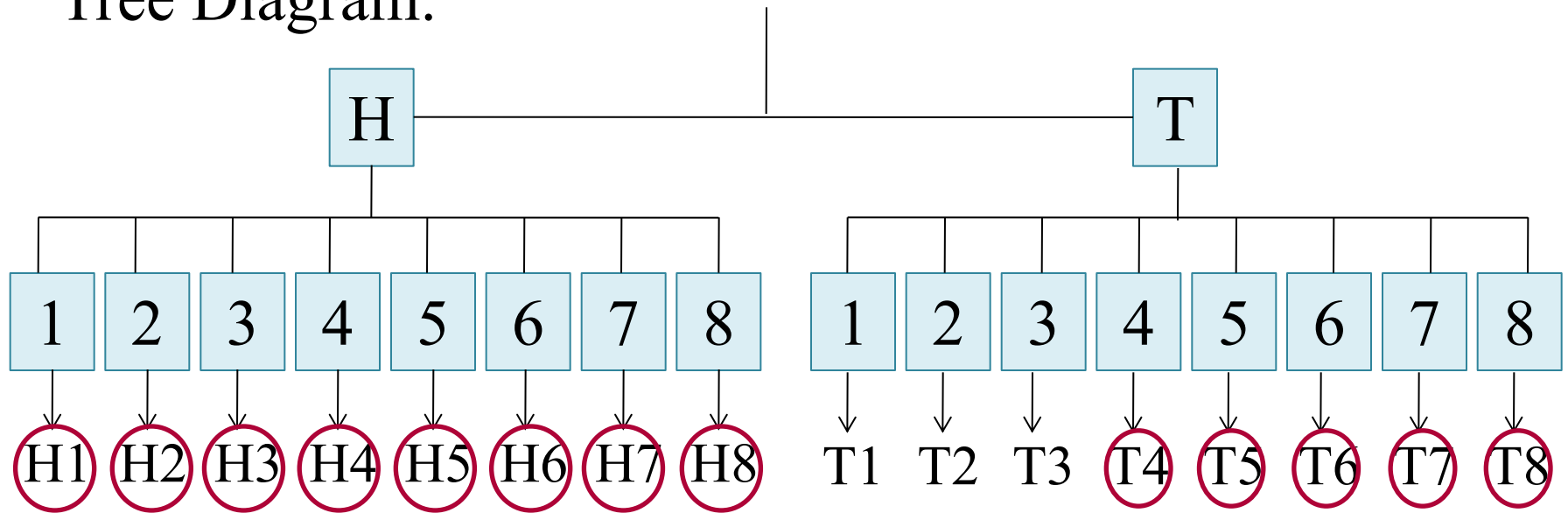


1. Event $A = \{T1, T3, T5, T7\}$

$$P(\text{tossing a tail and spinning an odd number}) = \frac{4}{16} = \frac{1}{4} = 0.25$$

Solution: Probability Using a Tree Diagram

Tree Diagram:



2. Event $B = \{H1, H2, H3, H4, H5, H6, H7, H8, T4, T5, T6, T7, T8\}$

$$P(\text{tossing a head or spinning a number greater than 3}) = \frac{13}{16} \approx 0.813$$

Example: Using the Fundamental Counting Principle

Your college identification number consists of eight digits. Each digit can be 0 through 9 and each digit can be repeated. What is the probability of getting your college identification number when randomly generating eight digits?



Solution: Using the Fundamental Counting Principle

- Each digit can be repeated
- There are 10 choices for each of the 8 digits
- Using the Fundamental Counting Principle, there are
$$10 \cdot 10 \cdot 10 \cdot 10 \cdot 10 \cdot 10 \cdot 10 \cdot 10$$
$$= 10^8 = 100,000,000 \text{ possible identification numbers}$$
- Only one of those numbers corresponds to your ID number

$$P(\text{your ID number}) = \frac{1}{100,000,000} \text{ or } 0.00000001$$